

OLS and Asymptotics

Plan for today

- OLS assumptions and unbiasedness: cross-sections and time series.
- Consistency and convergence in probability and mean square.
- The LLN, its conditions and OLS consistency.
- Convergence in distribution and the CLT.
- OLS asymptotic normality, inference and convergence rates.
- Course recap and the exam.

Appendix non-examinable

CMT, Slutsky, the delta method, stochastic order and vector limits.

Parameters, estimators and estimates

In Lecture 1, we distinguished three objects:

- A parameter such as β is a population target.
- An estimator such as $\hat{\beta}$ is a random rule.
- A realized estimate is one number from one sample.

As before, assumptions on the DGP let us establish properties of an estimator.

Today we ask what happens to $\hat{\beta}_n$ as the sample size grows.

Recall: regression notation

We use the same regression notation as in Lecture 3:

$$y = X\beta + \varepsilon.$$

Equivalently, observation by observation,

$$y_i = x_i'\beta + \varepsilon_i, \quad i = 1, \dots, n.$$

- x_i is a $k \times 1$ vector of regressors.
- X is the $n \times k$ matrix with row x_i' .

OLS will be our running example for consistency and asymptotic normality.

Cross-section: earnings and education

Suppose we observe earnings and education for n people in one year:

$$\log(\text{earnings}_i) = \beta_0 + \beta_1 \text{education}_i + \varepsilon_i, \quad i = 1, \dots, n.$$

- One observation is one **person**.
- $x_i = (1, \text{education}_i)'$; stacking the observations gives $y = X\beta + \varepsilon$.
- OLS uses this sample to estimate β_0 and β_1 .
- Whether OLS works well depends on how education and the unobserved factors in ε_i are related.

From assumptions to properties

If the DGP satisfies certain assumptions,
then the estimator has certain properties.

- The DGP describes how the regressors and errors generate our data.
- Unbiasedness, consistency and efficiency describe how well our estimator works across repeated samples.
- Our question is: which assumptions make OLS work well?

The same framework

- In a cross-section, i represents a person. In a time series, t represents a time period.
- In both cases, the model is $y = X\beta + \varepsilon$ and OLS is

$$\hat{\beta} = (X'X)^{-1}X'y.$$

- We still ask whether $\hat{\beta}$ is unbiased, consistent or efficient.
- For time-series consistency, a larger sample means observing more periods:
 $T \rightarrow \infty$.

Time series: consumption and income

Suppose we observe aggregate consumption and income each quarter:

$$\text{consumption}_t = \beta_0 + \beta_1 \text{income}_t + \varepsilon_t, \quad t = 1, \dots, T.$$

- One observation is one **quarter**.
- $x_t = (1, \text{income}_t)'$; stacking the observations again gives $y = X\beta + \varepsilon$.
- We use the same OLS formula to estimate β_0 and β_1 .
- Now shocks may persist across quarters, and income may be related to those shocks.

Time series: what changes?

There are two differences:

1. **Notation.** We move from people $i = 1, \dots, n$ to time periods $t = 1, \dots, T$.
2. **Assumptions.** Assumptions that hold in a cross-section may not hold in a time series. For example, observations across time may be dependent.

The logic is the same: DGP assumptions imply properties of the estimator.

When does OLS work well?

- A1: Linear model. $y = X\beta + \varepsilon$, with $\beta \in \mathbb{R}^k$.
- A2: Full column rank. $\text{rank}(X) = k$, so $X'X$ is invertible.
- A3: Conditional mean zero. $\mathbb{E}(\varepsilon \mid X) = 0$.
- A3': Zero regressor–error moments. A weaker alternative to A3:

$$\mathbb{E}(x_i \varepsilon_i) = 0 \quad (\text{or } \mathbb{E}(x_t \varepsilon_t) = 0 \text{ for time series}).$$

- A4: Homoscedasticity and no autocorrelation.

$$\text{Var}(\varepsilon \mid X) = \sigma^2 I_n, \quad 0 < \sigma^2 < \infty.$$

OLS: headline results

Unbiasedness

A1, A2, A3

$$\mathbb{E}(\hat{\beta}) = \beta.$$

Efficiency

A1, A2, A3, A4

OLS is BLUE.

Best linear unbiased
estimator.

Consistency

A1, A2, A3' + LLN

$$\hat{\beta}_n \xrightarrow{p} \beta.$$

OLS unbiasedness

OLS unbiasedness: A1–A3

Start from OLS and substitute the model:

$$\begin{aligned}\hat{\beta} &\stackrel{\text{A2}}{=} (X'X)^{-1}X'y \\ &\stackrel{\text{A1}}{=} (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= \beta + (X'X)^{-1}X'\varepsilon.\end{aligned}$$

Take expectations conditional on X :

$$\mathbb{E}(\hat{\beta} \mid X) = \beta + (X'X)^{-1}X'\mathbb{E}(\varepsilon \mid X) \stackrel{\text{A3}}{=} \beta.$$

Finally, apply the law of iterated expectations:

$$\mathbb{E}(\hat{\beta}) \stackrel{\text{LIE}}{=} \mathbb{E}\left[\mathbb{E}(\hat{\beta} \mid X)\right] = \mathbb{E}[\beta] = \beta.$$

A1–A3 are enough.

- Errors may be heteroscedastic or autocorrelated.
- A4 is needed for the Gauss–Markov efficiency result, not this proof.

Asymptotic properties

Recall: consistency

As the sample size grows, a consistent estimator approaches its target:

$$\hat{\beta}_n \xrightarrow{P} \beta.$$

- Repeated samples give a **sampling distribution** for each n .
- As n grows, it concentrates near β .
- The probability of being “far away” goes to zero.

OLS simulation setup

Repeated-sampling experiment:

$$y_i = 1 + 2x_i + \varepsilon_i.$$

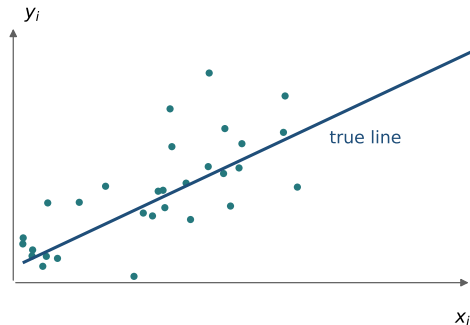
Here

$$x_i \sim N(0, 1), \quad u_i \sim t_5,$$

$$\varepsilon_i = 2\sqrt{3/5} u_i$$

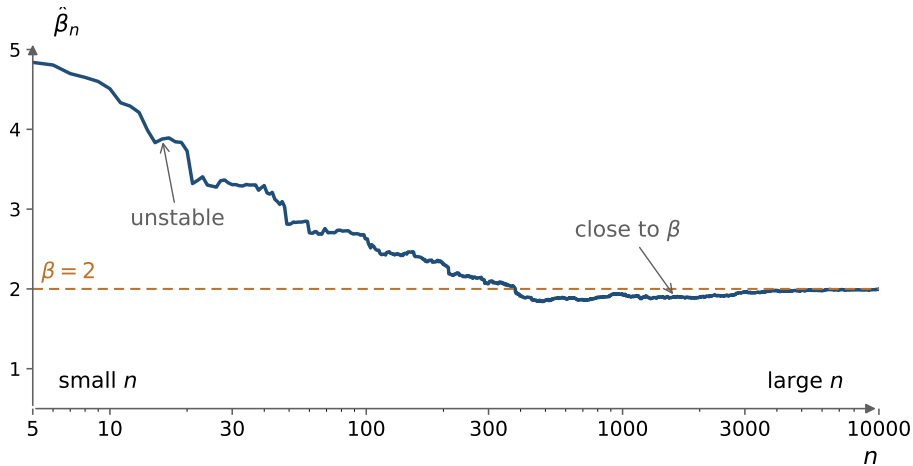
with iid observations and x_i independent of ε_i .

- The true slope is $\beta = 2$; the errors are fat-tailed with finite variance.
- We repeat the experiment 250,000 times at several sample sizes.



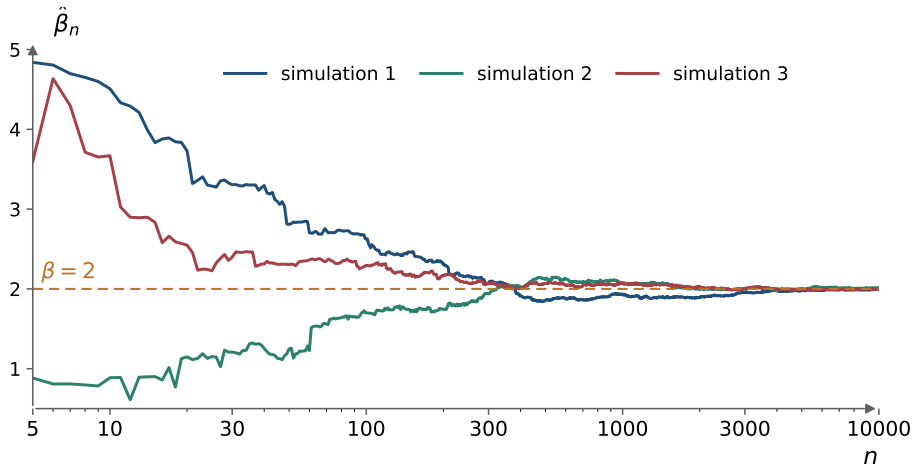
One sample and the population line

Consistency over n



One simulated OLS path

Consistency over n : three paths



Three simulated OLS paths

What does “far away” mean?

In Lecture 1, we built up to consistency by measuring distance from the target.

1. $|\hat{\beta}_n - \beta|$ is the distance between our estimator and the target.
2. Pick a fixed $\varepsilon > 0$. Call the estimator “far away” if $|\hat{\beta}_n - \beta| > \varepsilon$.
3. For example, $\varepsilon = 1/2$ means more than half a unit away from β .
4. Across repeated samples, the probability of being far away is

$$\Pr(|\hat{\beta}_n - \beta| > \varepsilon).$$

Consistency: however small the fixed distance, this probability goes to zero.

Convergence in probability

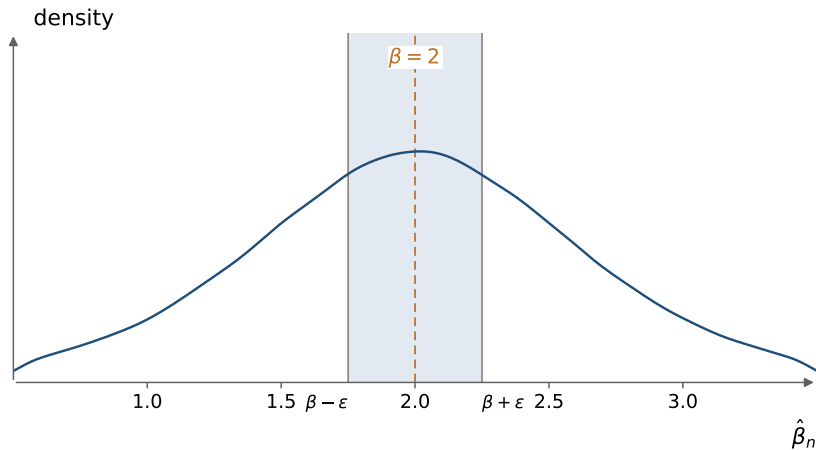
Writing that idea formally, $\hat{\beta}_n$ is **consistent** for β if

$$\Pr(|\hat{\beta}_n - \beta| > \varepsilon) \longrightarrow 0 \quad \text{as } n \rightarrow \infty, \quad \text{for every } \varepsilon > 0.$$

- We call this **convergence in probability** and write $\hat{\beta}_n \xrightarrow{P} \beta$.
- More generally, $Z_n \xrightarrow{P} z$ means the same thing with Z_n and z in place of the estimator and its target.
- Keep ε fixed as n grows: the probability outside $[\beta - \varepsilon, \beta + \varepsilon]$ goes to zero.

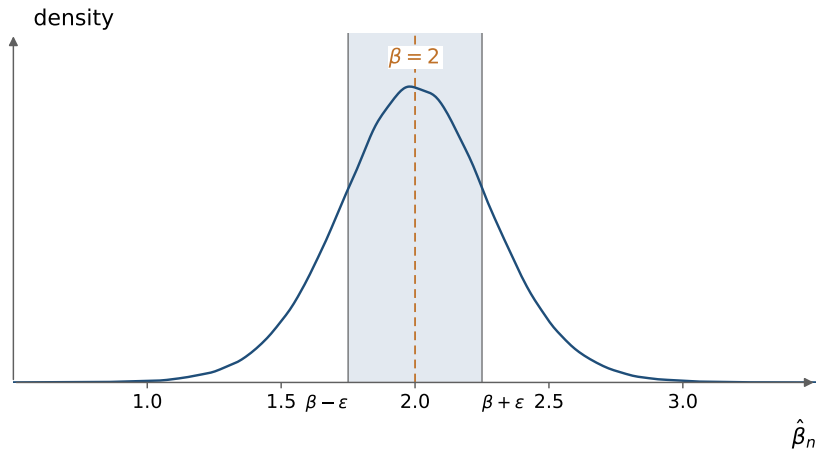
The next pictures show this probability shrinking for one fixed band.

$n = 10$



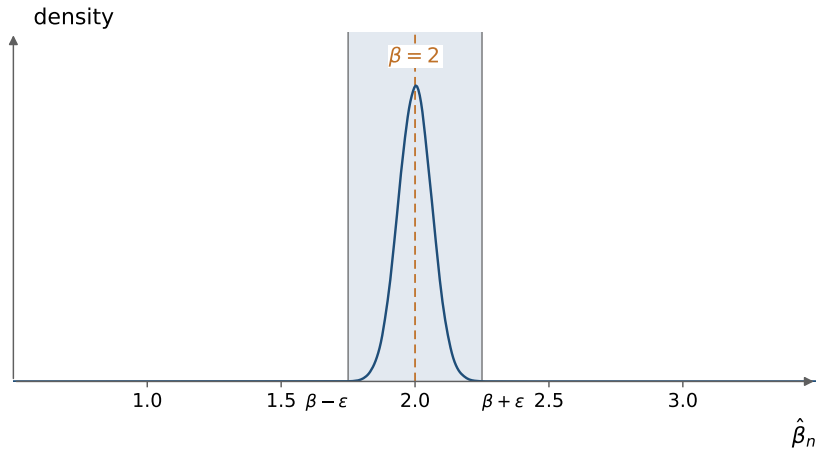
$$\Pr(|\hat{\beta}_n - \beta| > \varepsilon) \approx 0.70$$

$n = 50$



$$\Pr(|\hat{\beta}_n - \beta| > \epsilon) \approx 0.38$$

$n = 1000$



$$\Pr(|\hat{\beta}_n - \beta| > \varepsilon) \approx 0.00$$

Mean Square Convergence

Mean square convergence says the **expected squared error goes to zero**:

$$\mathbb{E}[(\hat{\beta}_n - \beta)^2] \longrightarrow 0.$$

For an estimator,

$$\text{MSE}(\hat{\beta}_n) = \mathbb{E}[(\hat{\beta}_n - \beta)^2] = \text{Bias}(\hat{\beta}_n)^2 + \text{Var}(\hat{\beta}_n).$$

- If bias $\rightarrow 0$ and variance $\rightarrow 0$, then MSE $\rightarrow 0$, which implies consistency.
- Consistency does not imply mean square convergence; **rare huge errors** can keep MSE large.

The Law of Large Numbers

Why do we need the LLN?

The LLN is the key to showing consistency.

- Estimators such as OLS depend on sample averages.
- The LLN tells us when those averages approach their population counterparts.
- First, we develop the LLN in general and ask when it holds.
- Then, we use it to show that OLS approaches the true coefficient.

The LLN: LSE student heights

Suppose we sample students at random from LSE and ask their height.

- The **population** is all LSE students; our **sample** is the students we ask.
- A small sample might happen to include unusually tall or short students.
- As n grows, the mix of heights in the sample tends to look more like the mix in the population.
- So the average height in the sample gets close to the average height in the population.

Sample average height $\xrightarrow{P}$ Population average height.

The LLN: intuition

As n grows, **sample averages** get close to **population averages**.

$$\frac{1}{n} \sum_{i=1}^n f(X_i) \xrightarrow{p} \mathbb{E}[f(X)].$$

We need conditions for this result:

iid observations and a **finite moment**: $\mathbb{E}|f(X_i)| < \infty$.

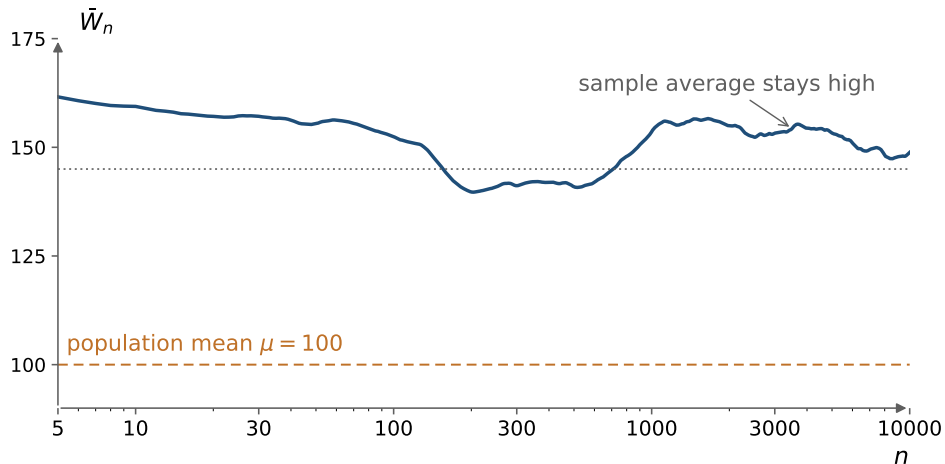
These are the conditions for this version of the LLN.

What if we sample people and then their friends?

Suppose we want to estimate average wealth in the population.

- We start with a few wealthy people, ask their wealth, then ask their friends, and their friends' friends.
- Wealthy people tend to have wealthy friends. Suppose our sample stays within this wealthy network.
- As n grows, the sample does not look more and more like the whole population.
- The sample average stays too high: more observations do not fix who we are sampling.

Friend sampling: one path



One simulated friend-sampling path

Changing distributions: theory

Suppose the observations are independent but become more variable over time.

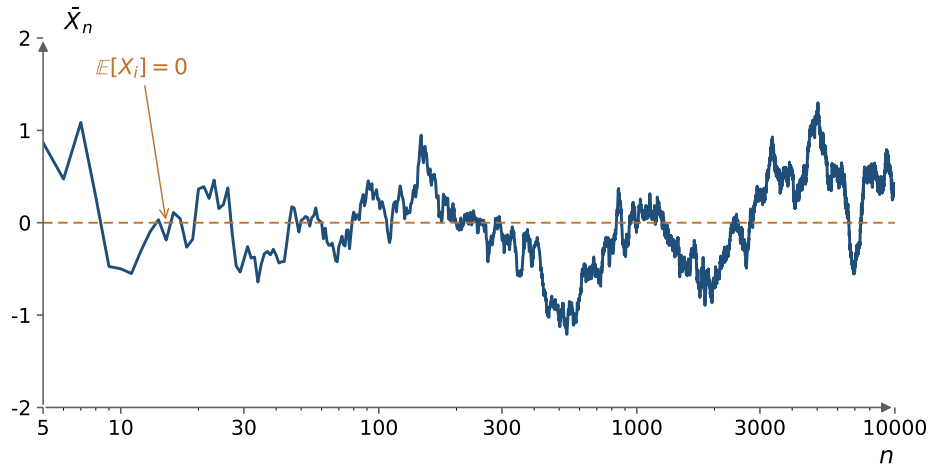
$$X_i \sim N(0, i), \quad \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n i = \frac{n+1}{2n} \rightarrow \frac{1}{2}.$$

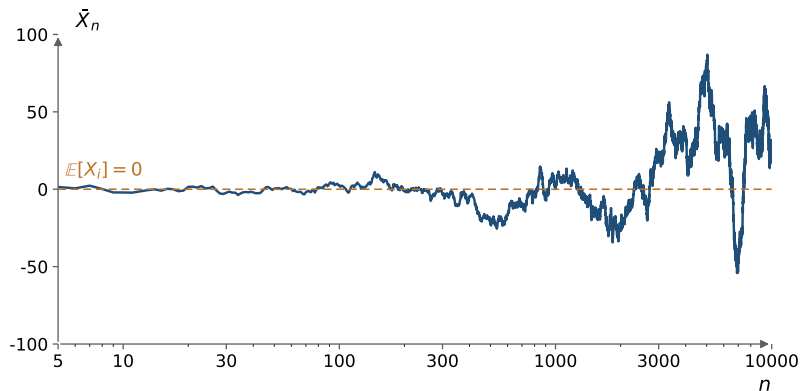
- Later observations are much noisier than earlier observations.
- The extra noise is not fully washed out by averaging.
- The average does not settle down around zero; its variance does not vanish.

Changing distributions: simulation



One simulated path for $X_i \sim N(0, i)$

A more severe drift: simulation



One simulated path for $X_i \sim N(0, i^2)$. Note that:

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n i^2 = \frac{(n+1)(2n+1)}{6n} \sim \frac{n}{3}.$$

What about time series?

Can we use the LLN with time-series data?

Observations across time need not be independent.

What replaces iid?

- The iid assumption does two jobs: observations are independent, and they have the same distribution.
- For time series, iid is usually too strong.
- We still need enough new information as time passes.
- We also need the distribution not to drift too much over time.
- The standard replacement is: **stationarity** plus **weak dependence**.
- With those assumptions, sample averages can still estimate population averages.

What is a distribution for a time-series variable?

Let X_t be inflation at date t .

- If we want to understand the effect of inflation on growth, we may run a regression connecting growth to inflation.
- Before a date arrives, inflation for that date is uncertain and is modelled as a random variable.
- One way to think about this is Bayesian: sitting in 1900, what would our beliefs have been about inflation in 2000?
- After 2000 arrives, we observe one realization of that random variable.

X_{2000} = one realized draw from the distribution of inflation in 2000.

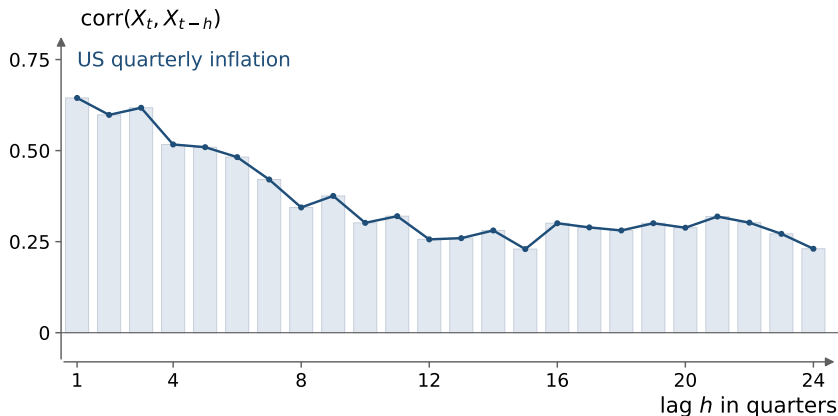
Stationarity

A common working definition is weak stationarity:

$$\mathbb{E}[X_t] = \mu, \quad \text{Var}(X_t) = \gamma_0, \quad \text{Cov}(X_t, X_{t-h}) = \gamma_h.$$

- The realized inflation rates can differ from year to year.
- The mean and variance **remain stable** over calendar time.
- Dependence between two observations is described by their **separation h** .
- This is the time-series version of “similar enough distributions.”
- Without it, old observations may be drawn from a different environment than new observations.

Weak dependence: inflation



US quarterly inflation, 1959–2009

- Nearby quarters are related, but the relationship weakens at longer lags.

LLN for a time series

A typical statement has the same form:

$$\frac{1}{T} \sum_{t=1}^T f(X_t) \xrightarrow{p} \mathbb{E}[f(X_t)].$$

- We replaced i by t and n by T .
- The assumptions changed from iid to stationarity and weak dependence.
- The conclusion is the same: sample averages estimate population averages.

OLS consistency

From two observations to X

Recall Lecture 3: an intercept and one regressor, with just two observations.

$$y_i = x_i' \beta + \varepsilon_i, \quad x_1 = \begin{pmatrix} 1 \\ z_1 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 1 \\ z_2 \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

Stack the two equations:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & z_1 \\ 1 & z_2 \end{pmatrix}}_{X: 2 \times 2} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \end{pmatrix}.$$

$$X = \begin{pmatrix} x_1' \\ x_2' \end{pmatrix}, \quad X' = \begin{pmatrix} 1 & 1 \\ z_1 & z_2 \end{pmatrix} = \begin{pmatrix} x_1 & x_2 \end{pmatrix}.$$

Each x_i' is a **row of X** ; each x_i is a **column of X'** .

Matrix blocks: $X'X$ as a sum

Each x_i is a $k \times 1$ column vector; row i of X is x_i' .

$$X'X = \underbrace{\begin{pmatrix} x_1 & x_2 & \cdots & x_n \end{pmatrix}}_{X': k \times n} \underbrace{\begin{pmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{pmatrix}}_{X: n \times k} = \underbrace{\sum_{i=1}^n x_i x_i'}_{k \times k}.$$

Multiply matching blocks, then add.

Left: $(k \times n)(n \times k)$ gives $k \times k$.

Right: each $x_i x_i'$ is $(k \times 1)(1 \times k)$; the sum is also $k \times k$.

Matrix blocks: $X'\varepsilon$ as a sum

Each ε_i is a scalar, so $x_i\varepsilon_i$ is a $k \times 1$ vector.

$$X'\varepsilon = \underbrace{\begin{pmatrix} x_1 & x_2 & \cdots & x_n \end{pmatrix}}_{X': k \times n} \underbrace{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}}_{\varepsilon: n \times 1} = \sum_{i=1}^n \underbrace{x_i \varepsilon_i}_{k \times 1}.$$

Again, multiply matching blocks, then add.

Left: $(k \times n)(n \times 1)$ gives $k \times 1$.

Right: each $x_i\varepsilon_i$ is a $k \times 1$ vector times a scalar; the sum is also $k \times 1$.

OLS consistency: sample averages

Start as in the unbiasedness proof:

$$\begin{aligned}\hat{\beta} &\stackrel{A2}{=} (X'X)^{-1}X'y \\ &\stackrel{A1}{=} (X'X)^{-1}X'(X\beta + \varepsilon) = \beta + (X'X)^{-1}X'\varepsilon.\end{aligned}$$

Now use the identities we have just established:

$$\begin{aligned}\hat{\beta} - \beta &= \left(\sum_{i=1}^n x_i x_i' \right)^{-1} \left(\sum_{i=1}^n x_i \varepsilon_i \right) \\ &= \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \right).\end{aligned}$$

The change of notation: creates sample averages

OLS consistency: apply the LLN

Assume the sample moments obey an LLN and $Q = \mathbb{E}[x_i x_i']$ is nonsingular.

$$\frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} Q, \quad \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{p} \mathbb{E}[x_i \varepsilon_i] \stackrel{A3'}{=} 0.$$

Because Q is nonsingular, the inverse of the first average converges to Q^{-1} .

$$\begin{aligned} \hat{\beta} - \beta &= \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \right) \\ &\xrightarrow{p} Q^{-1} \cdot 0 = 0. \end{aligned}$$

Therefore $\hat{\beta} \xrightarrow{p} \beta$: OLS is consistent.

Consistency: assumptions used

- A1 and A2 give the OLS decomposition.
- Writing $X'X$ and $X'\varepsilon$ as sums lets us apply the LLN.
- A3', $\mathbb{E}[x_i\varepsilon_i] = 0$, makes the error moment converge to zero.
- We need a nonsingular limiting matrix Q and technical rules for combining limits, developed further in other courses.

A3 and A4 are not needed for this proof.

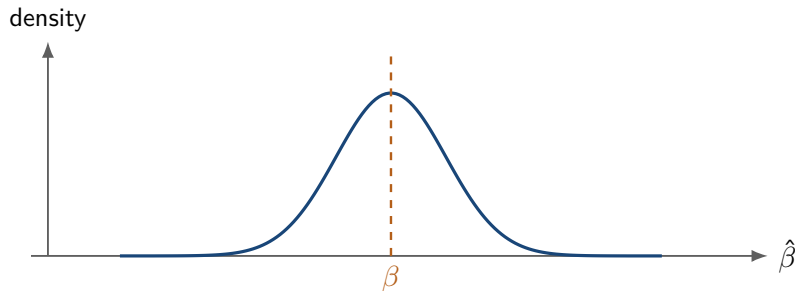
For time series, replace i, n by t, T and use suitable moment and dependence conditions for the LLN.

The Central Limit Theorem

The normal approximation

For a scalar coefficient, asymptotic theory often says

$$\hat{\beta} \stackrel{a}{\sim} N\left(\beta, \frac{V}{n}\right).$$



Rescaling the approximation

Start from the approximation

$$\hat{\beta} \stackrel{a}{\sim} N\left(\beta, \frac{V}{n}\right).$$

Subtracting β gives

$$\hat{\beta} - \beta \stackrel{a}{\sim} N\left(0, \frac{V}{n}\right).$$

Multiplying by $\sqrt{n}$ gives

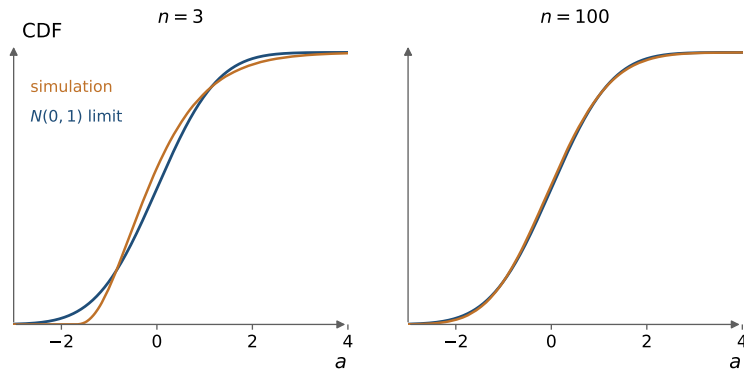
$$\sqrt{n}(\hat{\beta} - \beta) \stackrel{a}{\sim} N(0, V).$$

- These are the same approximation written in different units.
- The last form has a non-degenerate limiting distribution.

Convergence in distribution

Let $F_n(a) = \Pr(Z_n \leq a)$ and $F(a) = \Pr(Z \leq a)$.

$$Z_n \xrightarrow{d} Z \iff F_n(a) \longrightarrow F(a) \text{ for every continuity point } a \text{ of } F.$$



Consistency and asymptotic normality

- **Consistency:** as the sample grows, the estimator gets close to the target.

$$\hat{\beta}_n \xrightarrow{p} \beta.$$

- **Asymptotic normality:** after rescaling, the estimation error has a limiting normal distribution.

$$\sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} N(0, V).$$

Consistency tells us we approach the target.

Asymptotic normality helps us quantify the remaining uncertainty.

Lecture 4: approximate inference

- Showing that $\hat{\beta}$ is **asymptotically normal** justifies treating it as **approximately normal** in a large sample.
- With a suitable standard error, the T -statistic is then approximately standard normal under H_0 :

$$T = \frac{\hat{\beta} - \beta_0}{\text{se}(\hat{\beta})} \stackrel{a}{\sim} N(0, 1).$$

- This is the large-sample limit of the t -distribution, and the basis of the approximate hypothesis tests we used in **Lecture 4**.

The central limit theorem

- Suppose $X_1, \dots, X_n$ are iid, with mean μ and variance $0 < \sigma^2 < \infty$.
- The CLT describes the sampling error in the average:

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right) \xrightarrow{d} N(0, \sigma^2).$$

- In a large sample, we therefore use the approximation

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \stackrel{a}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right).$$

The observations need not be normal for their average to be approximately normal.

OLS: preparing for the CLT

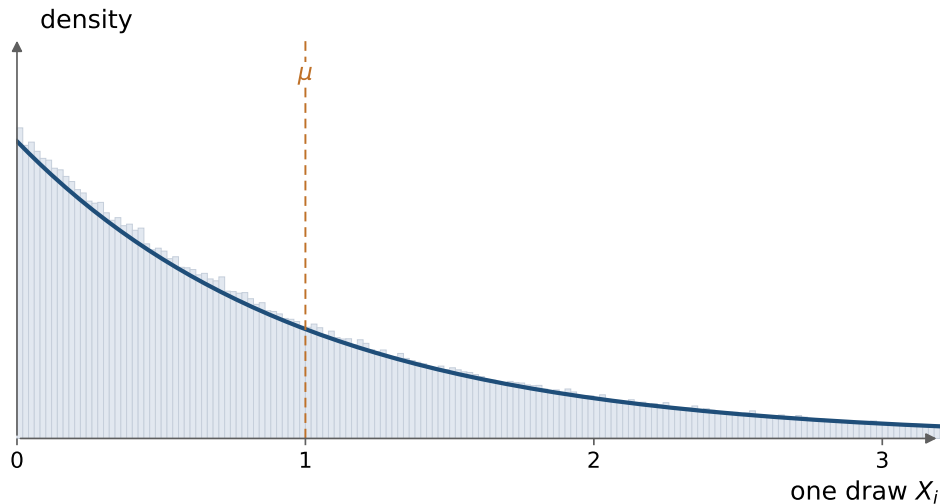
Under A1 and A2, we have already shown that

$$\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon.$$

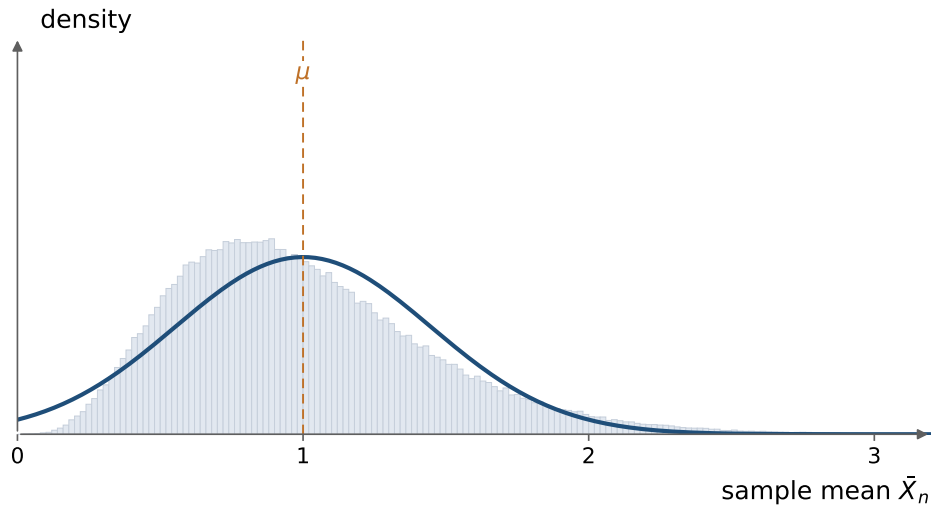
Write the matrix products as sums and multiply by $\sqrt{n}$:

$$\begin{aligned}\sqrt{n}(\hat{\beta} - \beta) &= \sqrt{n} \left(\sum_{i=1}^n x_i x_i' \right)^{-1} \sum_{i=1}^n x_i \varepsilon_i \\ &= \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \\ &= \underbrace{\left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1}}_{\text{we want to use an LLN}} \underbrace{\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i \right)}_{\text{we want to use a CLT}}.\end{aligned}$$

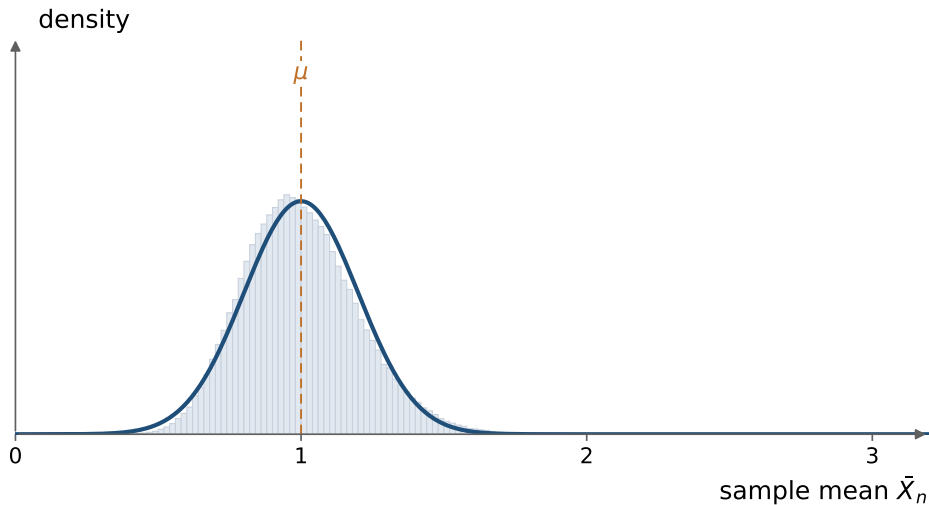
CLT simulation: exponential data



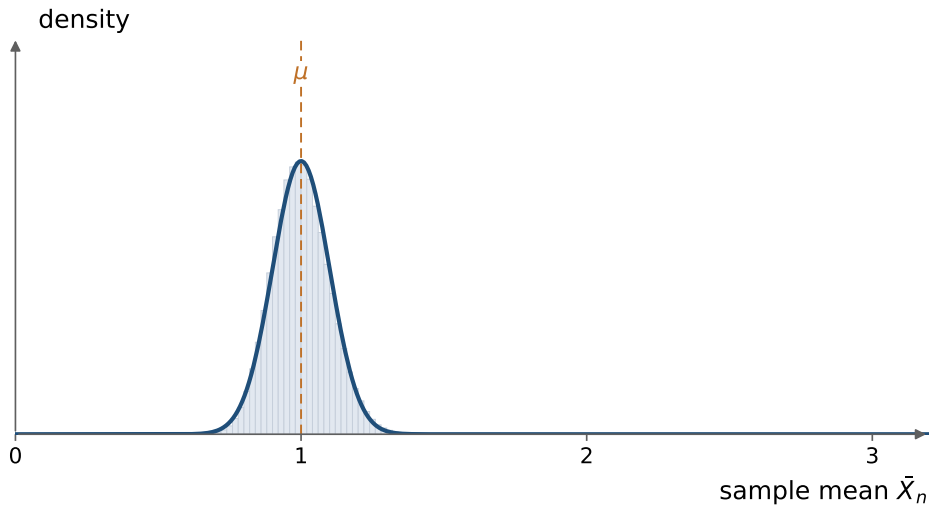
CLT simulation: average of 5 draws



CLT simulation: average of 25 draws



CLT simulation: average of 100 draws



Why a normal distribution?

Recall from Lecture 3: linear combinations of independent normals are normal.

$$X_1, X_2 \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2) \implies \frac{X_1 + X_2}{2} \sim N\left(\mu, \frac{\sigma^2}{2}\right).$$

Averaging keeps the normal shape, but shrinks the spread.

After rescaling, the normal is a “fixed point”:

$$Z_1, Z_2 \stackrel{\text{iid}}{\sim} N(0, 1) \implies \frac{Z_1 + Z_2}{\sqrt{2}} \sim N(0, 1).$$

This gives intuition for the stable shape we see in the CLT.

The CLT: what about time series?

- The classic CLT we just used assumes iid observations and finite variance.
- Time-series observations can be dependent, so we need a CLT that allows for this.
- We need stronger conditions than for the LLN: suitable stationarity, moment and weak-dependence conditions.
- Mixing conditions are one way to control that dependence.

Serial dependence also changes the variance of the limiting distribution.

OLS asymptotic normality

OLS normality: setup

Assume A1, A2 and A3', with iid observations. Also require

$$Q = \mathbb{E}[x_i x_i'] \text{ positive definite,} \quad \mathbb{E}\|x_i\|^2 < \infty, \quad \mathbb{E}\|x_i \varepsilon_i\|^2 < \infty.$$

Start from OLS and substitute the model:

$$\begin{aligned} \hat{\beta} &\stackrel{\text{A2}}{=} (X'X)^{-1} X'y \\ &\stackrel{\text{A1}}{=} (X'X)^{-1} X'(X\beta + \varepsilon) = \beta + (X'X)^{-1} X'\varepsilon. \end{aligned}$$

Using the matrix identities and multiplying by $\sqrt{n}$,

$$\sqrt{n}(\hat{\beta} - \beta) = \underbrace{\left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1}}_{\text{LLN}} \underbrace{\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i \right)}_{\text{CLT}}.$$

OLS normality: combine the limits

By A3', $\mathbb{E}[x_i \varepsilon_i] = 0$, so $S = \text{Var}(x_i \varepsilon_i) = \mathbb{E}[x_i x_i' \varepsilon_i^2]$.

The iid and moment conditions give

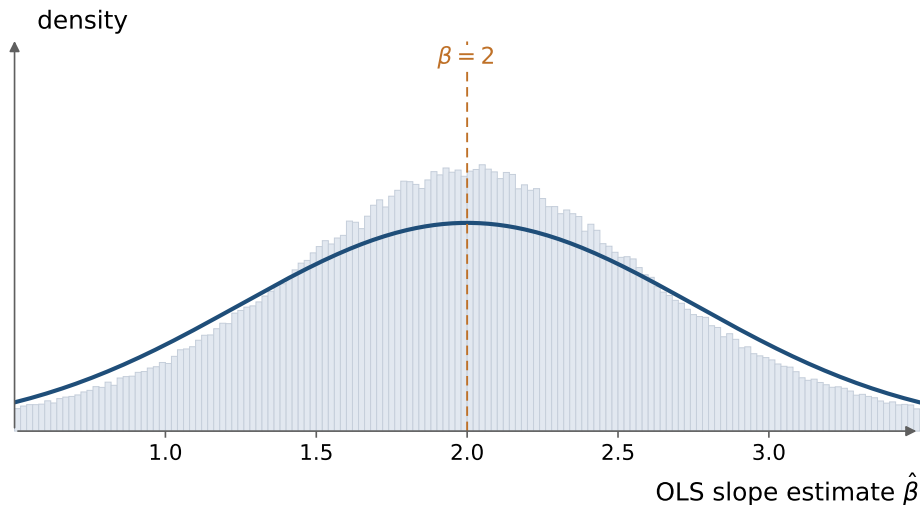
$$\frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} Q, \quad \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{d} Z \sim N(0, S).$$

With Q nonsingular,

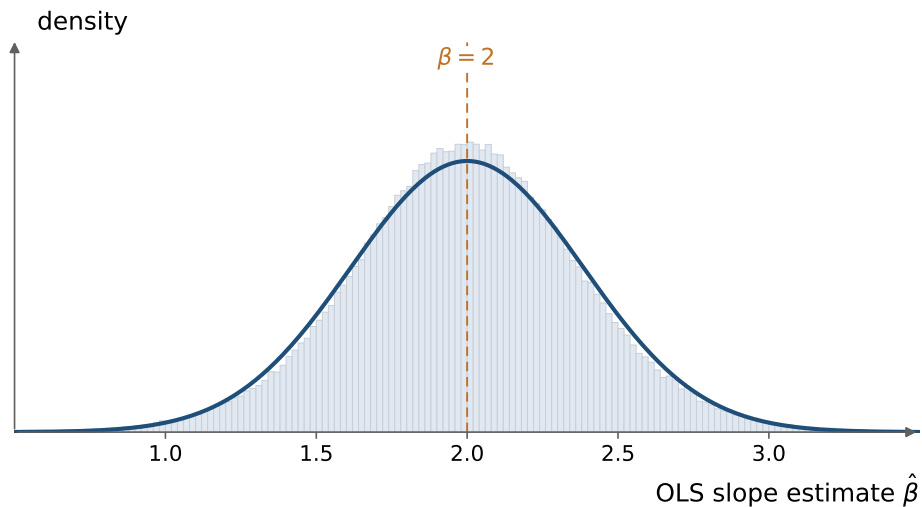
$$\begin{aligned} \sqrt{n}(\hat{\beta} - \beta) &= \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i \right) \\ &\xrightarrow{d} Q^{-1} Z \sim N \left(0, \underbrace{Q^{-1} S Q^{-1}}_V \right). \end{aligned}$$

OLS is asymptotically normal. A4 was not needed.

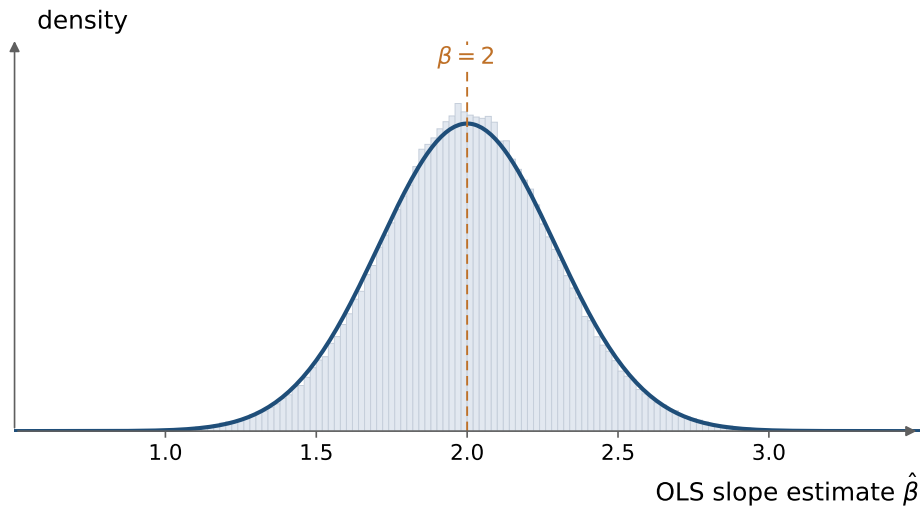
OLS sampling distribution: $n = 10$



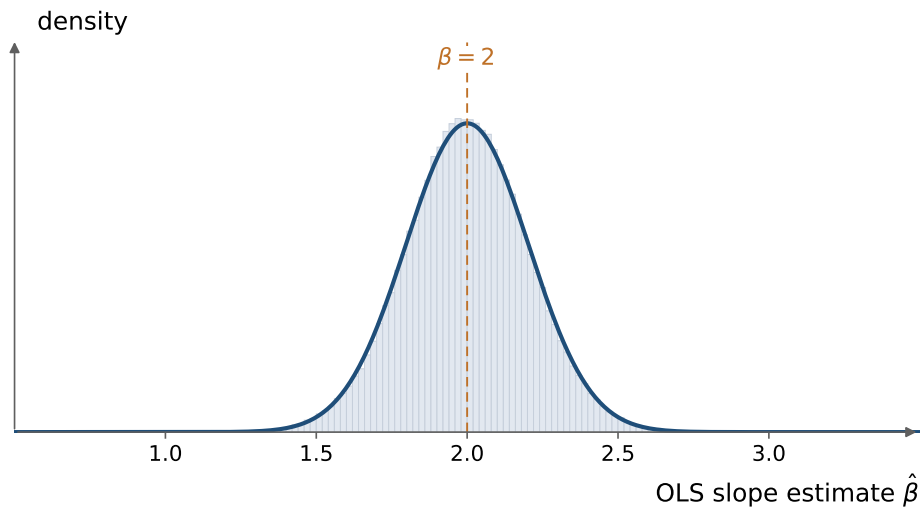
OLS sampling distribution: $n = 30$



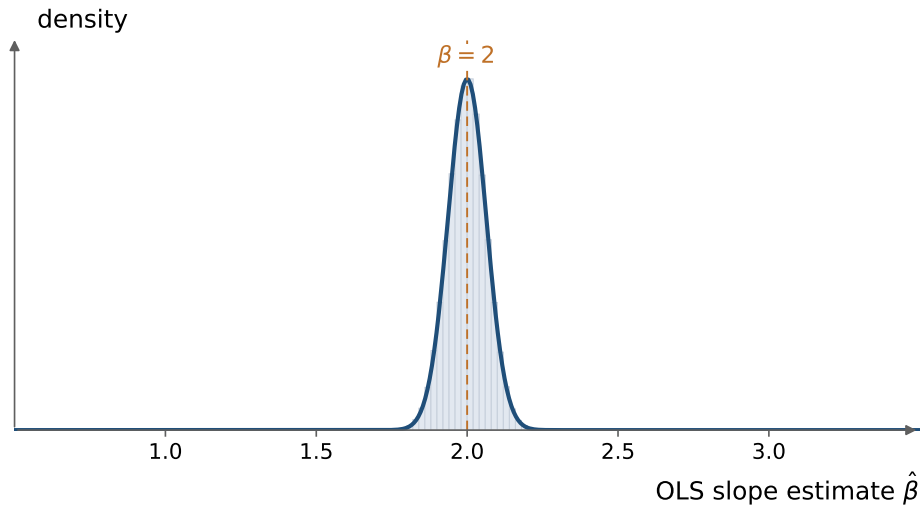
OLS sampling distribution: $n = 50$



OLS sampling distribution: $n = 100$



OLS sampling distribution: $n = 1000$



A battle of two forces

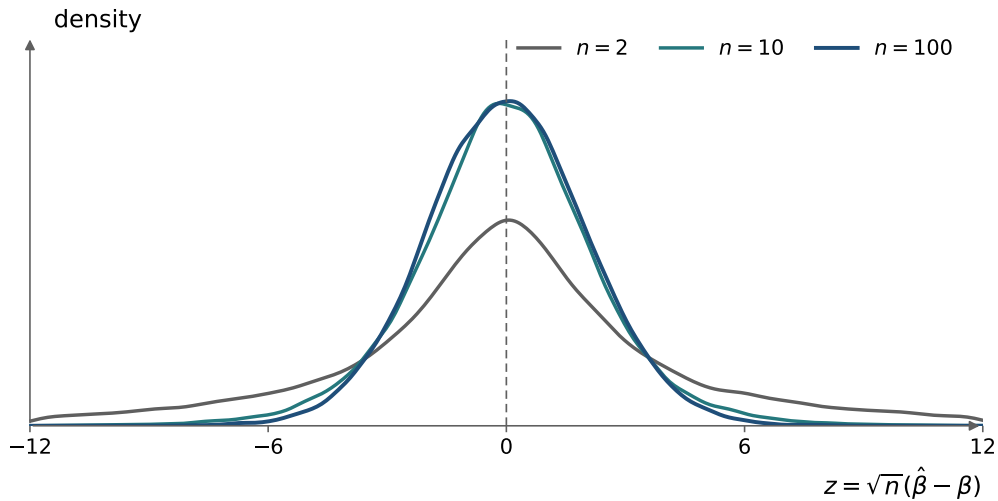
Two forces act in opposite directions:

$$\underbrace{\sqrt{n}}_{\text{grows}} \times \underbrace{(\hat{\beta}_n - \beta)}_{\text{shrinks in probability}} \xrightarrow{d} \underbrace{N(0, V)}_{\text{stable limiting distribution}} .$$

- More data make the estimation error smaller.
- Multiplying by $\sqrt{n}$ magnifies that shrinking error.
- The two forces balance: the scaled error has a stable distribution.

Asymptotic normality implies consistency.

A stable scaled distribution



Convergence rate

The same formula also gives the rough rate:

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V) \quad \implies \quad \hat{\beta} - \beta \text{ is roughly of size } \frac{1}{\sqrt{n}}.$$

- If the amount of data quadruples, the typical distance between $\hat{\beta}$ and β **roughly halves**.
- Finite-sample performance still depends on the data-generating process and the sample at hand.

Putting the ideas together

- OLS gives a concrete example of why we need convergence results.
- LLNs deliver **consistency**: sample moments get close to population moments.
- CLTs deliver **approximate distributions**: centered and scaled estimators have stable limits.
- The same proof pattern appears for IV and many other estimators.
- The final output is usable inference: standard errors, tests and confidence intervals.

Key messages

- Time series and panels keep the same estimation logic; the assumptions change.
- Consistency means convergence in probability; LLNs turn sample averages into population averages.
- CLTs turn scaled estimation errors into approximate normal distributions.
- Asymptotic normality implies consistency and gives the $n^{-1/2}$ convergence rate.

Summary

Lecture 1: Foundations

- **Identification:** connect the question to a population target.
- **Estimation:** learn about that target from a sample.
- Sampling distributions describe uncertainty; estimator properties depend on DGP assumptions.

Lecture 2: Statistics I

- Distributions describe uncertainty; moments summarise it.
- Conditioning links joint and marginal distributions.
- Independence, mean independence and zero correlation differ.

Lecture 3: Statistics II

- Normal vectors connect to χ^2 , t and F distributions.
- Tails determine moments; curvature matters for expectations.
- Matrices organise regression and error covariance.

Lecture 4: Hypothesis Testing

- Test whether the data are surprising under H_0 .
- Critical values, p -values and confidence intervals give linked decisions.
- Size controls false rejections; power measures detection.

Lecture 5: OLS and Asymptotics

- OLS properties follow from assumptions on the DGP.
- The LLN gives consistency; the CLT gives approximate inference.
- Time series uses the same algebra, with different dependence conditions.

The exam

- Exam is on Tuesday 22nd in the morning. Remedial sessions if that doesn't go well in AT.
- Examinable content = what I talk about in lectures + any concepts explicitly introduced in the seminar. **Nothing else.**
- 2 sample exams on moodle

In conclusion

- You should now have all the foundations you need for an MSc/MRes course in statistics or metrics.
- Please feel free to refer back to any of the resources during the year, including appendices.
- And say hello if you see me around campus!

Appendix: Extra tools

non-examinable

Extra tools: roadmap non-examinable

- CMT and Slutsky: familiar limit rules for random variables.
- Delta method: approximate distributions for functions of estimators.
- O_p and o_p : shorthand for how large random quantities are.
- A brief look at vector limits.

For a continuous function g , ordinary limits satisfy

$$x_n \rightarrow x \implies g(x_n) \rightarrow g(x).$$

CMT gives the same rule for random variables:

$$X_n \xrightarrow{p} X \implies g(X_n) \xrightarrow{p} g(X),$$

$$X_n \xrightarrow{d} X \implies g(X_n) \xrightarrow{d} g(X).$$

Slutsky in the OLS proof non-examinable

Addition and multiplication are continuous. For constant A ,

$$A_n \xrightarrow{p} A, \quad B_n \xrightarrow{d} B \quad \implies \quad A_n B_n \xrightarrow{d} AB, \quad A_n + B_n \xrightarrow{d} A + B.$$

In our OLS proof:

$$\sqrt{n}(\hat{\beta} - \beta) = \underbrace{\left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1}}_{\xrightarrow{p} Q^{-1}} \underbrace{\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i \right)}_{\xrightarrow{d} Z \sim N(0, S)}$$

Slutsky

$$\xrightarrow{d} Q^{-1} Z \sim N(0, Q^{-1} S Q^{-1}).$$

Delta method: proof intuition non-examinable

Suppose $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V)$.

Near β , Taylor's theorem says a smooth function is almost a straight line:

$$g(\hat{\beta}) - g(\beta) \approx g'(\beta)(\hat{\beta} - \beta).$$

Multiply both sides by $\sqrt{n}$:

$$\sqrt{n}(g(\hat{\beta}) - g(\beta)) \approx g'(\beta) \underbrace{\sqrt{n}(\hat{\beta} - \beta)}_{\xrightarrow{d} N(0, V)}.$$

- The Taylor remainder vanishes in probability even after multiplying by $\sqrt{n}$.
- Multiplying a normal by $g'(\beta)$ multiplies its variance by $[g'(\beta)]^2$.

$$\sqrt{n}(g(\hat{\beta}) - g(\beta)) \xrightarrow{d} N(0, [g'(\beta)]^2 V).$$

For a scalar β , suppose

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V).$$

If $g : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at β , then

$$\sqrt{n}(g(\hat{\beta}) - g(\beta)) \xrightarrow{d} N(0, [g'(\beta)]^2 V).$$

A shorthand for the “of size” language we used earlier:

$$X_n = O_p(a_n)$$

X_n/a_n is bounded in probability.

An upper bound on stochastic size.

$$X_n = o_p(a_n)$$

$$X_n/a_n \xrightarrow{p} 0.$$

Negligible relative to a_n .

For an asymptotically normal estimator,

$$\hat{\beta} - \beta = O_p(n^{-1/2}), \quad \hat{\beta} - \beta = o_p(1).$$

Two further signposts non-examinable

- Cramér–Wold: vector convergence in distribution can be checked through all scalar linear combinations.

$$Z_n \xrightarrow{d} Z \iff \lambda' Z_n \xrightarrow{d} \lambda' Z \text{ for all } \lambda.$$

- Slutsky-type convergence rules: combine a limiting distribution with a constant probability limit.

$$Z_n \xrightarrow{d} Z, \quad A_n \xrightarrow{p} A \implies A_n Z_n \xrightarrow{d} AZ.$$

- These tools recur in econometrics and are developed in later courses.